Biogas-Expert: Sustainable biomethane production in Northern-Germany - Nitrogen leaching after application of biogas residue

Svoboda N.[1] , Wienforth B.[2] , Sieling K.[2] , Kage H.[2] , Taube F.[1] , Herrmann A.[1]

Abstract

To ensure the largest possible benefit of replacing fossil fuels by bioenergy, it is mandatory to produce bioenergy in a sustainable way. With respect to biomethane production there is limited data on its risk of contaminating the groundwater with nitrogen. A two year trial located in two landscapes in Northern Germany was conducted to analyse the N leaching potential of biogas residue compared to other N fertiliser types for monocropped maize. Nitrogen load was obtained from measured N concentration in leachate and simulated soil water flow. Significant differences in N load among the different N fertiliser types were detected, which could be attributed mainly to the mineral share of N input.

Keywords: methane, biogas residue, nitrate leaching, maize

Introduction

The production of methane from anaerobic digestion of slurry and/or biomass has greatly expanded in Germany due to substantial subsidisation (> 4000 plants in 2009). While initially regarded favourable, criticism has been voiced recently due to fuel-food competition and potential environmental impact. Biogas residue, which is produced in large amounts, should be used in a sustainable way. The objective of this study was to assess the N leaching potential of biogas residue applied to maize compared to animal manure and mineral fertiliser.

Materials and methods

A study was conducted through a 2-year field experiment (2007-2009) on two experimental stations (Hohenschulen, Karkendamm) of Kiel University. Hohenschulen is characterised by an average precipitation of 750 mm and a mean annual temperature of 8.3°C. Soil type is a luvisol with 17% clay in the topsoil (sandy loam). Karkendamm has an average precipitation of 844 mm and a mean annual temperature of 8.6°C. Soil type can be classified as a gleyic Podzol with

[1]Christian-Albrechts-University of Kiel, D-24118 Kiel; Institute of Crop Science and Plant Breeding; Grass and Forage Science/Organic Agriculture

[2]Christian-Albrechts-University of Kiel, D-24118 Kiel; Institute of Crop Science and Plant Breeding; Agronomy and Crop Science

less than 1% clay in the topsoil (sandy sand). Weather conditions in the experimental years differed from the long term values. At Karkendamm, mean annual temperature (10.3°C in 2007, 9.7°C in 2008) was above long-term average (8.6°C), while rainfall was slightly above long-term average in 2007 (898 mm), and somewhat below in 2008 (726 mm). At Hohenschulen, temperature and precipitation in 2007 was substantially higher (926 mm; 10.1°C) compared to long term conditions, while 2008 was slightly drier (722) and warmer (9.8°C). Maize (cv. Ronaldino) was planted mid till end of April. N fertiliser was applied as mineral N (calcium ammonium nitrate), cattle slurry (Karkendamm only), pig slurry (Hohenschulen only), and biogas residue (cofermentation) in 4 levels (0, 120, 240, 360 kg N ha-1), split into 2 equal dressings.

Leachate was sampled weekly from April 2007 till March 2009 using ceramic suction cups (P80), installed 60 cm below ground, while soil moisture content was determined by TDRprobes. N load in the leachate was obtained from measured N concentrations and simulated leachate amount. Soil water balance and plant growth were simulated using the object oriented model library HUME within the Delphi© Builder programming environment (Kage H. and Stützel H., 1999). The model uses the Penman-Monteith equation for calculating the potential evapotranspiration. Plant growth was calculated by fitting simple functions to crop height and leaf area index (obtained by Licor LAI2000 instrument). Simulation of soil water movement was based on soil water potential. Relationship between soil water diffusivity and the volumetric water content was described using the functions of van Genuchten as revised by Woesten J.H.M. and van Genuchten, M.T. (1988). The required parameters were estimated based on field measurement; additionally empirical data were used. The relation of N input to N load in leachate was analysed by SAS Proc GLM assuming a quadratic function, where N input was considered as covariable. Data were analysed separately for each site and year since monitoring periods differed due to missing values caused by technical reasons.

Results and Discussion

It is generally assumed that the use of residues from biogas production in crop production would increase yield and consequently decrease the N leaching losses because of their higher NH4-N content compared to animal slurries. To test this hypothesis we compared the N load in the leachate after application of biogas residues with application of mineral fertiliser, cattle slurry and pig slurry. The biogas residues used in the present study were somewhat untypical since they had lower NH4-N content than cattle slurry and pig slurry (Table 1).

	$kgNm^{-3}$	$\%NH_4-N$	$kgNm^{-3}$	$\%NH_4-N$	$kgNm^{-3}$	$\%NH_4-N$
Karkendamm 2007	3.1	54.1	-	-	3.7	51.0
Karkendamm 2008	2.9	64.6	-	-	4.0	53.7
Hohenschulen 2007	-	-	3.7	73.3	3.7	51.0
Hohenschulen 2008	-	-	4.2	65.3	4.0	53.7

Table 1: Nitrogen ($kgNm^{-3}$) and ammonia (% NH4-N) contents of cattle slurry, pig slurry and biogas residue applied to maize. Cattle slurry Pig slurry Biogas residue

The model calibration for the soil water balance resulted in a satisfactory agreement between observed and calculated soil water contents (R^2 0.1-0.6; RMSE 0.03-0.06 cm^3cm^{-3}), which allowed to calculate the N loads. The major form of nitrogen in leachate was nitrate, accounting for at least 94% of total N. The statistical analysis refers only to periods, where measurements of N concentration in the leachate were available for all treatments, see Figure 1. The analysis of covariance using total N input as covariable showed a significant impact of fertiliser type, N input (quadratic term), and interaction between fertiliser type and N input (linear term) on N amount in leachate (not presented). As expected, a significantly higher N load was found for mineral N, while organic fertilisers had similar N losses. When the mineral share of N input was used as covariable, fertiliser type and the interaction between fertiliser type and N input no longer had an effect on N load, which confirms the hypothesis, that the mineral share of N input explains most of the variation among the fertiliser types. This is surprising when taking into account that biogas residue produced higher ammonia emission than cattle slurry and pig slurry after field application (Gericke D., 2010), while maize yield was similar (not shown). Comparable studies on the N leaching potential of biogas residues are scarce. Pötsch E. (2005) reported significantly lower nitrate concentration in leachate for biogas residue compared to mineral N fertiliser applied on grassland. A lysimeter study on the effect of N losses in wheat reported differences between biogas residue and animal manure depending on the type of soil used (Sächsische Landesanstalt, 1999), with biogas residue causing lower N losses on a loamy Chernozem. Brenner A. and Clemens J. (2005) detected less nitrate loss compared to unfermented slurry under cereals in one out of three leaching periods.

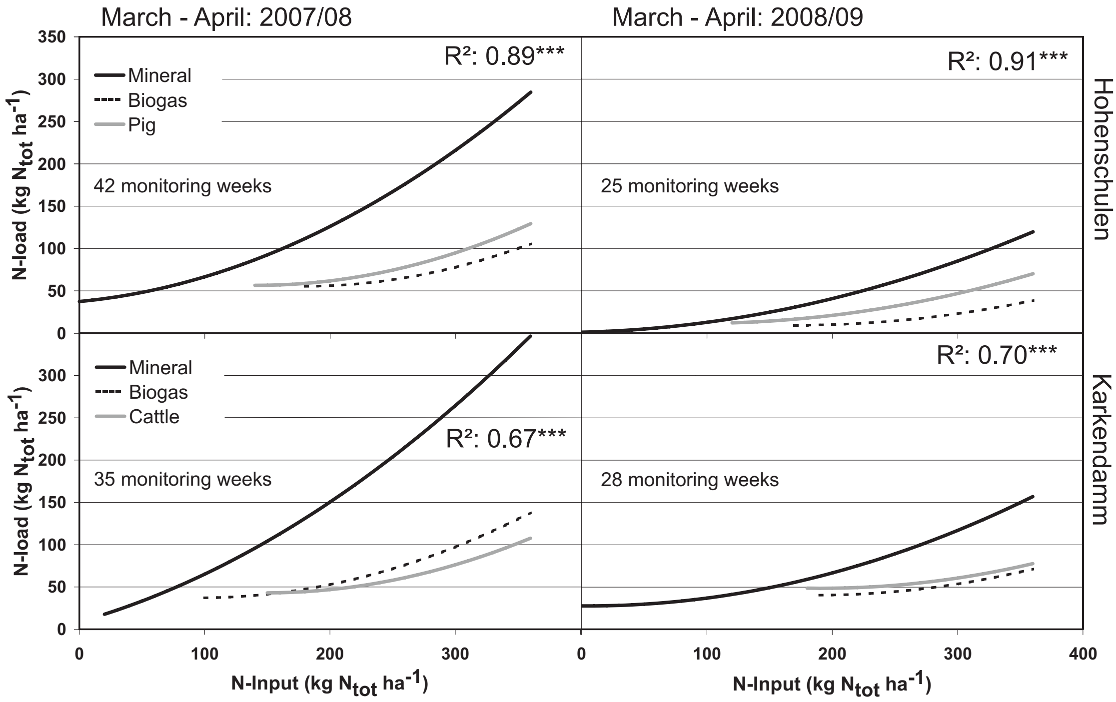

Figure 1: Relationship of total N input ($kgNha^{-1}$) and nitrogen load in leachate ($kgNha^{-1}$) as influenced by fertiliser type.

Conclusion

Application of biogas residue resulted in N losses comparable to animal manure. Potential N losses, however, are underestimated since the monitoring periods did not cover complete years. Therefore, the N balance will be simulated in a next step to allow N loss calculations over the whole 2-year period.

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